

S7 B.TECH – CSE

23CSE401 Fundamentals of Artificial Intelligence

Unit 1





ARTIFICIAL INTELLIGENCE

From History & Foundations to Intelligent Agents

Search Strategies · Problem Characteristics · Agent Systems · Ethics · Games

Topics: History of AI | Search Algorithms | Problem Solving | Intelligent Agents | Ethics | Games & Puzzles

Based on Russell & Norvig, Luger, and multiple authoritative AI sources

SECTION 01

History & Foundations of AI

From early dreams to modern intelligence

What is Artificial Intelligence?

AI is the simulation of human intelligence processes by machines — especially computer systems — including learning, reasoning, problem-solving, perception, and language understanding.

Learning

Acquiring data and rules
from experience

Reasoning

Using rules to reach
conclusions

Problem Solving

Searching for optimal
solutions

Perception

Interpreting sensory input

WHAT IS ARTIFICIAL INTELLIGENCE?

Artificial Intelligence (AI) is the science of making machines that can think, learn, reason, and act. It combines computer science, mathematics, psychology, linguistics, and philosophy.

THINK HUMANLY
Cognitive Simulation Approach

Systems that model human thought. Alan Newell & Herbert Simon's General Problem Solver (1961) mimicked step-by-step human reasoning. Basis of cognitive science and computational neuroscience.

ACT HUMANLY
Turing Test — Alan Turing,
1950

"Computing Machinery and Intelligence" — if a machine's responses are indistinguishable from a human's in text conversation, it is intelligent. ELIZA (1966) was an early example that fooled many users.

THINK RATIONALLY
Laws of Thought — Logic &
Reasoning

Use formal symbolic logic (Aristotle's syllogisms → Prolog). "If Socrates is a man and all men are mortal, then Socrates is mortal." Challenge: Not all knowledge fits neatly into formal logic.

ACT RATIONALLY ★
Rational Agent — Russell &
Norvig (AIMA)

Most widely accepted approach. An agent acts to achieve the BEST expected outcome given available information. Does not require human-like processes — just best possible action. Basis of modern AI.



What is Artificial Intelligence?

01

Act Like Humans

The Turing Test

System passes if human can't distinguish it from a person

02

Think Like Humans

The Cognitive Model

Process must mirror human reasoning — not just results

03

Think Rationally

Laws of Thought

Logical deduction — drawing valid conclusions from facts

04

Act Rationally

Rational Agent

Reaching correct conclusions via best possible actions

HISTORY OF ARTIFICIAL INTELLIGENCE — KEY MILESTONES

AI did not emerge overnight — it is the result of decades of research, breakthroughs, and painful setbacks (called 'AI Winters'). Understanding this history helps you appreciate WHERE modern AI comes from.

1943

Neural Nets Born:

McCulloch & Pitts publish first mathematical model of a neuron — the mathematical foundation that every modern neural network still builds on.

1950

Turing Test:

Alan Turing asks 'Can machines think?' in 'Computing Machinery and Intelligence' — proposes the Imitation Game as a test of machine intelligence.

1956

AI is Born:

Dartmouth Conference: John McCarthy, Marvin Minsky, Claude Shannon coin 'Artificial Intelligence'. AI becomes an official academic field.

1970s–80s

AI Winters:

Unrealistic promises lead to funding cuts. Research nearly stops. Expert systems (MYCIN for medicine, XCON for computers) keep the field alive during lean years.

1997

Deep Blue Wins:

IBM's chess computer defeats world champion Garry Kasparov. First time a machine beats a human champion in a complex game — massive public milestone.

2012+

Deep Learning Era:

AlexNet wins ImageNet with deep CNNs. GPT, AlphaGo, DALL-E follow. AI enters everyday life: smartphones, healthcare, autonomous vehicles, chatbots.

AI SYSTEM CHARACTERISTICS — WHAT MAKES A SYSTEM 'INTELLIGENT'?

An AI system is not defined by a single feature. True intelligence requires ALL of these working together:

REASONING

Drawing conclusions from known facts. Deductive: 'All birds have wings; eagle is a bird → eagle has wings.' Inductive: 'Every swan I've seen is white → all swans may be white.' AI systems use logic engines and knowledge bases.

LEARNING

Acquiring knowledge from experience WITHOUT being explicitly programmed for every case. A spam filter starts dumb but learns from thousands of flagged emails. Types: Supervised, Unsupervised, Reinforcement Learning.

PROBLEM SOLVING

Finding a sequence of actions that leads from a problem state to a goal state. Examples: Navigating GPS routes, scheduling exams, solving the Rubik's Cube. Uses search, planning, and constraint satisfaction.

PERCEPTION

Interpreting raw sensory data (pixels, audio waves, text) into meaningful information. Examples: Face unlock on your phone, Alexa hearing 'play music', a medical AI reading X-rays to detect cancer.

LANGUAGE

Understanding and generating human language. Natural Language Processing (NLP) powers Google Search, machine translation, chatbots, and voice assistants. Challenge: language is ambiguous, context-dependent, and constantly evolving.



Common Characteristics of Intelligent Systems

Regardless of domain, all intelligent systems share these five core characteristics:



Symbolic Information

Process abstract facts, concepts, and deduce conclusions



Domain Knowledge

Must know the problem environment to find solutions



Incomplete / Uncertain Data

Handle noisy, inaccurate, or conflicting information



Heuristic Methods

Guide search without guaranteeing optimality; find 'good enough' solutions



Adaptive Behavior

Change strategy when the environment changes

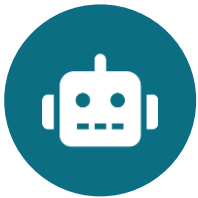
SECTION 02

Introduction to AI Systems

Types, branches, tools, and real-world applications



Types of AI: Narrow vs General



NARROW AI

(Weak AI)

- AI that surrounds us today
- Designed for ONE specific task
- No general consciousness
- Example: Google Translate — excellent at translation, cannot drive a car



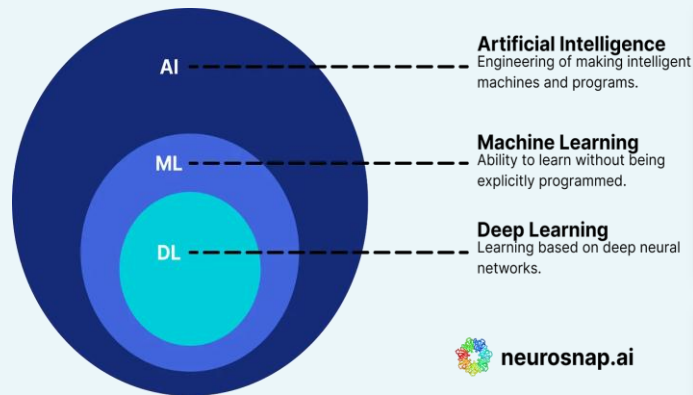
GENERAL AI

(Strong AI — Theoretical)

- Understands ANY problem a human can
- Reasoning + emotion + creativity
- Comparable to human intelligence
- Long-term research goal — not yet achieved



Key Branches of Artificial Intelligence



Machine Learning

Systems learn from data automatically — no explicit programming for every task



Computer Vision

Machines interpret visual data — object detection, facial recognition, medical imaging



Deep Learning

Multi-layer neural networks inspired by the brain — excels at complex pattern recognition



Natural Language Processing

Enables computers to read, understand, and generate human language meaningfully



Expert Systems

Rule-based AI using domain knowledge; central to 1980s AI — still used in law & medicine



Robotics

AI + mechanical engineering — automate physical tasks from factory floors to surgery





Real-World Applications of AI



Healthcare

MRI/X-ray analysis, early cancer detection, drug discovery



Finance

Fraud detection, algorithmic trading, risk assessment



Transport

Self-driving cars using computer vision + deep learning + sensor fusion



E-Commerce

Collaborative filtering finds 'people who bought X also bought Y'; neural nets model complex user-product relationships to drive sales.

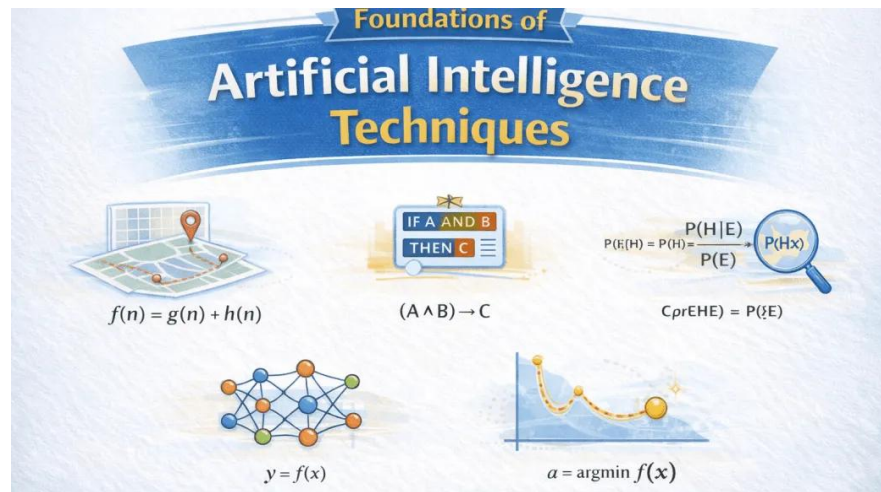
Amazon/Netflix recommendations via collaborative filtering & neural nets

Classification of Artificial Intelligence Techniques

- *Artificial Intelligence is often misunderstood as a single algorithm or a single technology.*
- *In reality, AI is a collection of carefully developed methodologies that address different types of problems.*

The foundations of AI can be viewed as five pillars:

- 1. Search** → How to find solutions.
- 2. Knowledge Representation** → How to store knowledge.
- 3. Reasoning under Uncertainty** → How to make decisions with incomplete information.
- 4. Learning** → How to improve from data.
- 5. Optimization & Reinforcement Learning** → How to select the best action.



“Structured problem-solving methodologies that enable systems to search, reason, learn, decide, and optimize.”

01

Search

Systematic exploration of a state space.

02

Symbolic

Reason over explicit knowledge with logic.

03

Probabilistic

Model uncertainty; update beliefs with evidence.

04

Learning

Discover patterns from data or experience.

05

Optimization

Find the best configuration among many.



Classification of AI Techniques

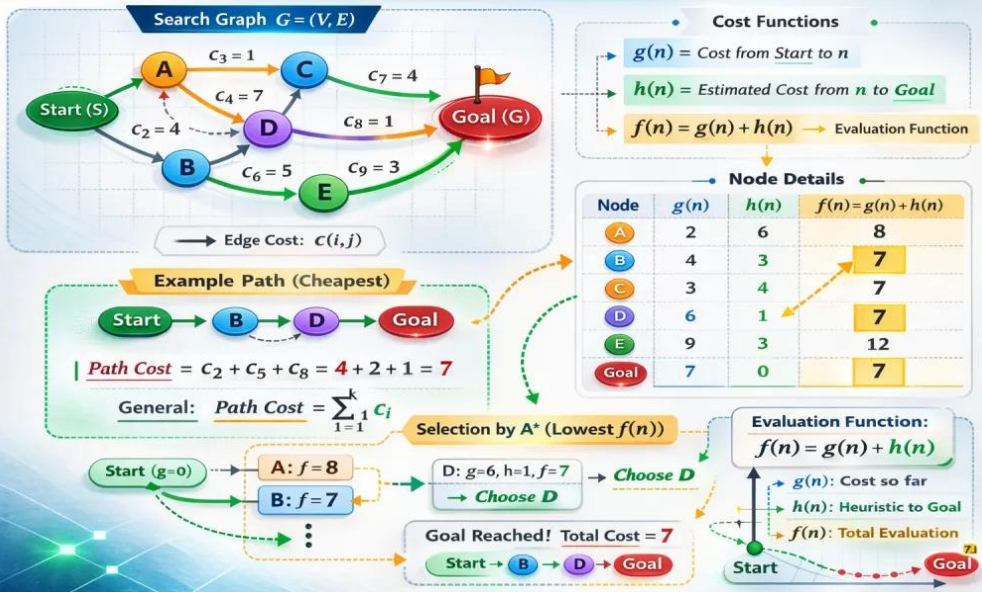
Technique Family	When to Use	Example Applications
Search-Based	Clear start, goal, and actions	GPS navigation, game playing
Knowledge-Based (Symbolic)	Explicit rules and structured knowledge	Medical expert systems, legal reasoning
Probabilistic	Uncertain or noisy environments	Spam detection, speech recognition
Learning-Based	Large datasets, hidden patterns	Face recognition, NLP, recommendations
Optimization & Evolutionary	Find best configuration among many	Scheduling, parameter tuning, robotics

Real AI systems combine multiple techniques — a self-driving car uses ALL five families simultaneously.

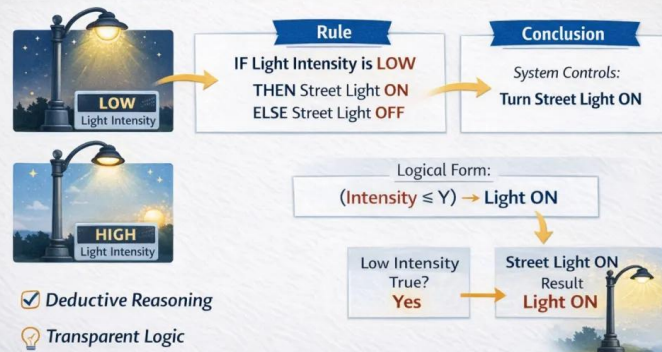


Search-Based, Knowledge-Based (Symbolic) & Probabilistic Techniques

Search Algorithms in Pathfinding (A*)

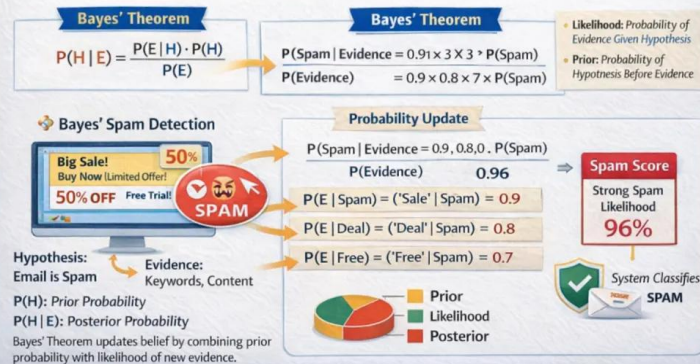


Knowledge-Based Reasoning Example



Probabilistic Techniques Example

Probability Quantifies Uncertainty: If H is a hypothesis and E is evidence, then the conditional probability $P(H|E)$ expresses the likelihood of the hypothesis given the observed evidence.





Learning-Based , Optimization & Evolutionary Techniques

Learning-Based Techniques in AI

Supervised Learning



House Price Prediction

Regression or Classification:
Learns from labeled examples
to map inputs to outputs

Unsupervised Learning



Clusters form based on common
attributes

Reinforcement Learning



Agent-Based Reward Maximization

Learns by interacting with an environment
to maximize a cumulative reward

Types of Learning in AI



Supervised Learning

Learns from labeled data
to predict outcomes



Unsupervised Learning

Finds patterns in
unlabeled data



Reinforcement Learning

Rewards agent
based on its decisions

Optimization and Evaluation Techniques

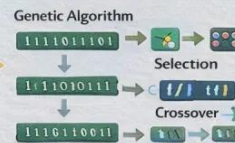
Gradient-Based Optimization



Uses gradients ($\nabla f(x)$) to iteratively
minimize a loss function

Example: Training neural networks

Evolutionary Optimization



Uses concepts of evolution
(selection, crossover, mutation) to
find the best solution

Example: Scheduling and resource
allocation

Hyperparameter Tuning



Searches for the best parameter
configuration of a model

Example: Tuning a machine learning
model

Evaluation Metrics



Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Learns from labeled data
to predict outcomes



Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Sensitivity in finding
Positives

F1-Score

$$\text{F1-Score} = 2 \left(\frac{\text{Prec.} \times \text{Rec.}}{\text{Prec.} + \text{Rec.}} \right)$$

Harmonic Mean of
Precision, Recall



MSE

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n y_i^2$$

Average of squared errors



Evaluates Model Quality
& Performance



SECTION 03

Problem Characteristics

Classifying AI problems for better problem-solving design

Before choosing a strategy, understand the **nature of the problem** — six dimensions determine which AI approach fits.



Problem Solving as Search

“Search-based AI formulates problem solving as the systematic exploration of a **state space** .” A delivery robot starting at A and reaching B at minimum cost is the canonical picture.

INITIAL STATE

Where the agent starts.

ACTIONS

Moves available in a state.

TRANSITION

How an action changes the state.

GOAL TEST

Is this state a solution?

PATH COST

Total cost of a sequence.

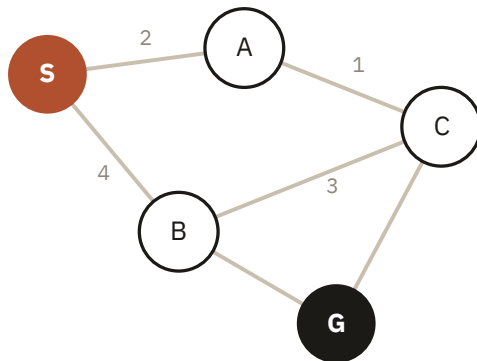
Formally — a problem is the tuple $\langle \text{states, initial state, actions, transition model, goal test, path cost} \rangle$. A *solution* is an action sequence leading from the initial state to a goal state.

States are **nodes (V)**; transitions are **edges (E)**, each with a cost c . The cost of a path is the sum of its edge costs.

```
G = (V, E)
```

```
PathCost =  $c_1 + c_2 + \dots + c_k$ 
```

```
 $g(n)$  = cost from start  $\rightarrow$  node  $n$ 
```



Tree vs. graph — graph search records visited states to avoid re-expanding them; tree search does not. The choice trades memory for protection against loops and redundant work.



Problem Formulation: State Space

A problem is defined by: Initial State, Possible Actions (Operators), Transition Model, Goal Test, and Path Cost

State

A snapshot of the world at a specific moment — encodes all relevant information about the current situation

Action / Operator

A valid move from one state to another. The Branching Factor = number of actions available per state

State Space

The complete set of all states + all valid actions — forms a graph $G = (V, E)$

Solution

A path through the state space graph from Initial State to a state satisfying the Goal Function

Path Cost

The total cost of a sequence of actions — we seek the minimum-cost path when optimizing

State Space Representation

Every AI problem maps to a **state space** — a graph of all possible situations connected by actions. AI's job: find the best path from start to goal.

4 Components of a State Space

Initial State

Where the agent begins.
Example: In(Kochi)

Goal State(s)

Desired end configuration.
Example: In(Munnar)

Actions / Operators

Legal moves between states.
Example: Drive(Kochi, Aluva)

Path Cost $g(n)$

Cost accumulated along a path. Example: Distance in km / travel time

Worked Example: Kerala Route Planning

Goal: Travel from Kochi to Munnar via the shortest road route

- **Initial State:** In(Kochi)
- **Goal State:** In(Munnar)
- **Actions:** Drive to any connected city
- **Path Cost:** Road distance (km)

 **Optimal A Path:***
Kochi → Aluva → Perumbavoor → Kothamangalam → Munnar
Total: ~130 km

 Smart search finds the best route without exploring every possible road combination.

Possible State Space:

Kochi → Aluva → Perumbavoor → Kothamangalam → Munnar



Problem Characteristics: Key Dimensions

To choose the most appropriate method, analyze the problem along these six key dimensions:

1

Decomposability

Can it be broken into independent sub-problems?

2

Ignorability

Can bad solution steps be ignored, undone, or are they permanent?

3

Predictability

Is the problem universe deterministic or uncertain?

4

Solution Type

Any-path solution acceptable, or must it be the BEST path?

5

Knowledge Role

How much domain knowledge is needed?

6

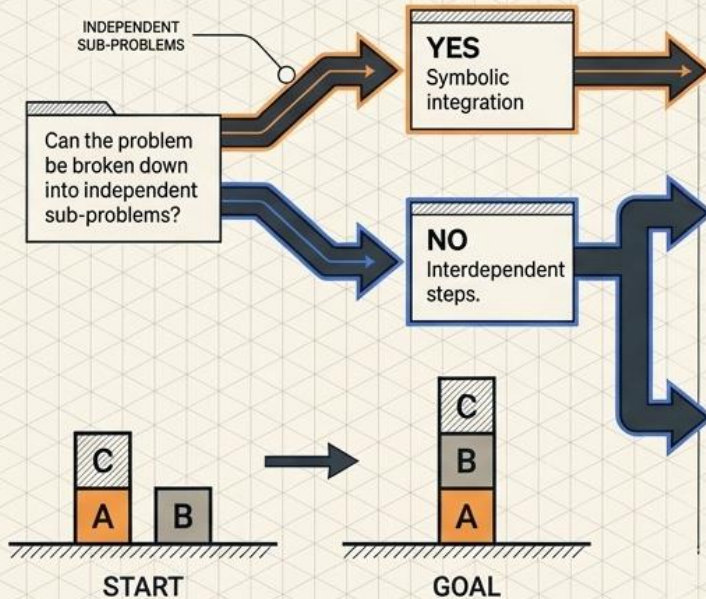
Interaction

Is human interaction required during solving?

PROBLEM CHARACTERISTICS & DIMENSIONS

DIMENSION 1: DECOMPOSABILITY

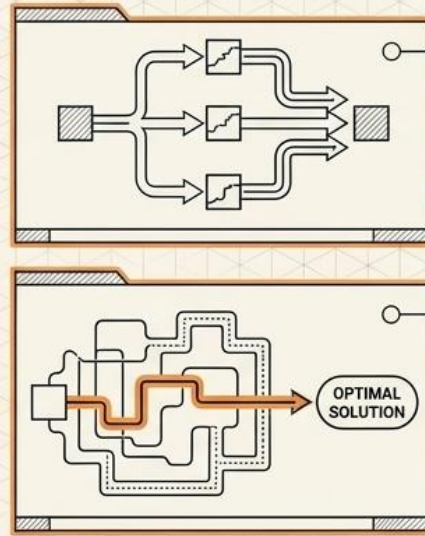
Can the problem be broken down into independent sub-problems?



DIMENSION 2: SOLUTION PATHS

Any Path: Satisfied with any valid solution (often uses heuristic methods).

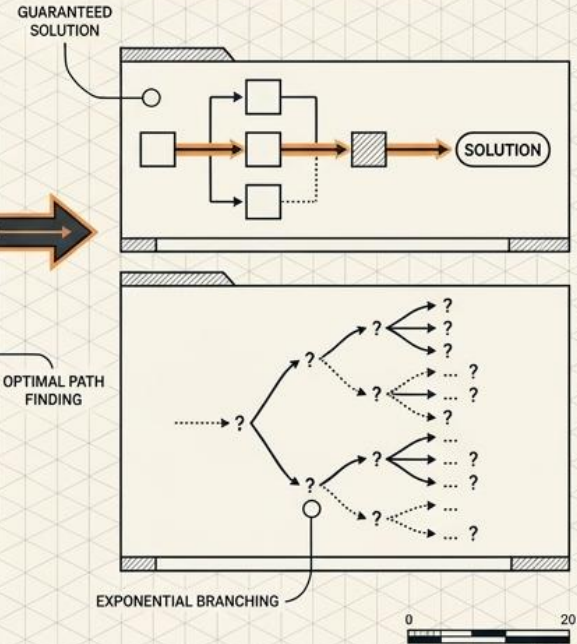
Best Path: Requires optimal solution via exhaustive search (computationally harder).



DIMENSION 3: OUTCOMES

Predictable: Planning guarantees a solution.

Uncertain: Search paths explode exponentially as outcomes cannot be guaranteed.

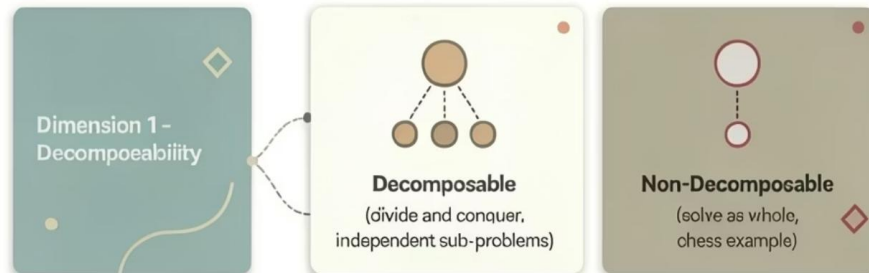


Problem Characteristics — Part 1 of 2

1. Decomposability

Can it be split into sub-problems?

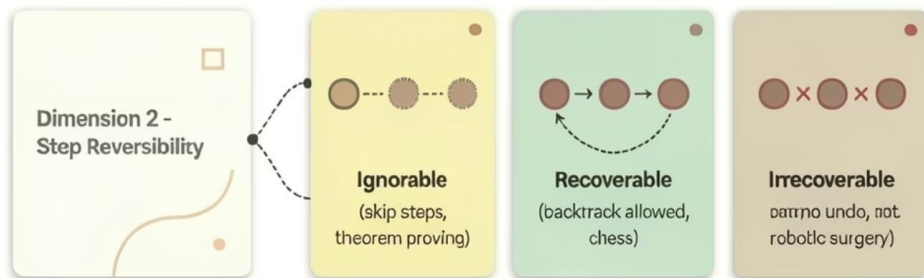
- **Yes** → Divide & conquer (e.g., solving equations independently)
- **No** → Solve as a whole (e.g., chess — one move affects the entire board)



2. Ignorable vs. Recoverable Steps

Can a wrong step be undone?

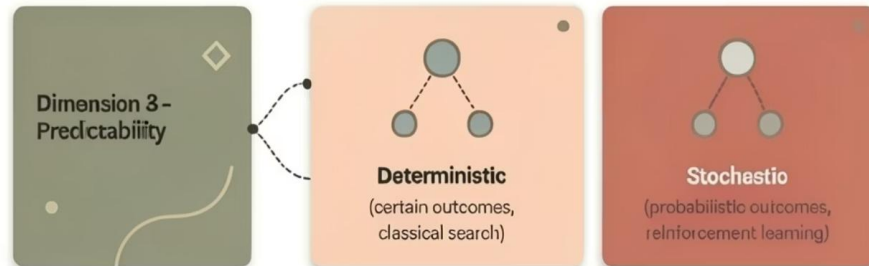
- **Ignorable:** Theorem proving — skip unhelpful axioms
- **Recoverable:** Chess — backtrack (costly but possible)
- **Irrecoverable:** Robotic surgery — a cut cannot be undone



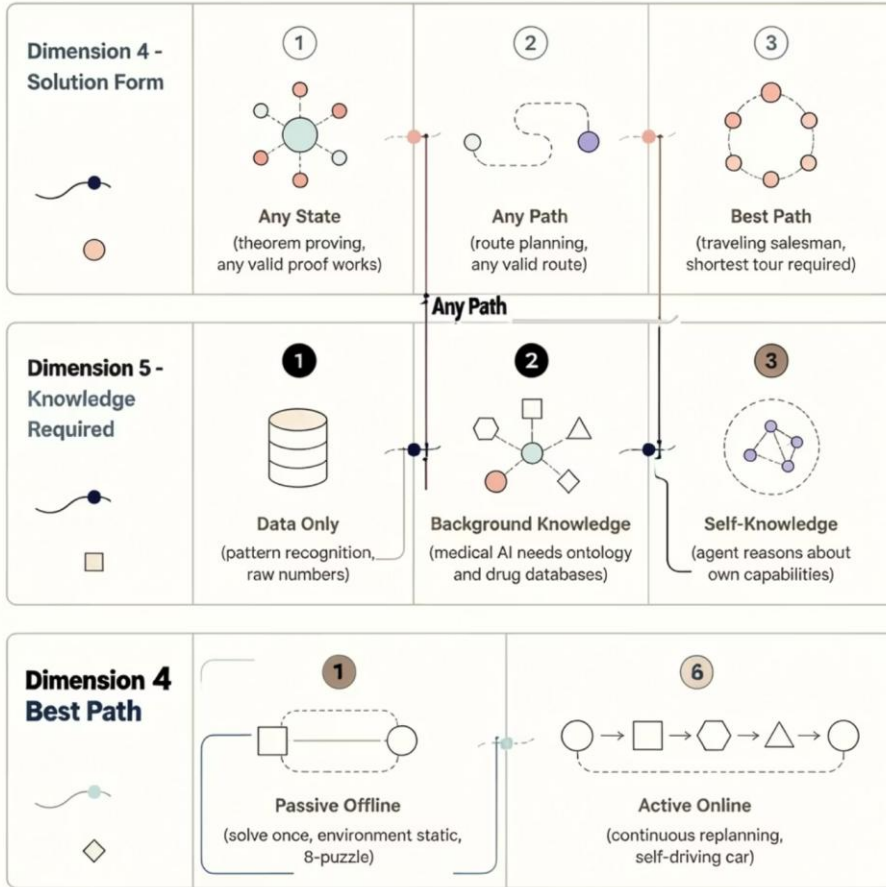
3. Predictability

Is the outcome of every action certain?

- **Deterministic:** 8-puzzle — same action always yields same result
- **Stochastic:** Weather forecasting, stock trading — need probability models



Problem Characteristics — Part 2 of 2



4. Solution Form

What does a valid solution look like?

- **Any state:** Theorem proving — any valid proof works
- **Any path:** Route from A to B — any valid route
- **Best path:** Traveling Salesman — must find the shortest tour

5. Knowledge Required

How much domain knowledge is needed?

- **Data-only:** Pattern recognition from raw numbers
- **Background knowledge:** Medical AI needs ontologies and drug databases
- **Self-knowledge:** Agent reasons about its own capabilities

6. Role of the Agent

Passive/Offline: Solve once (8-puzzle). **Active/Online:** Re-plan constantly (self-driving car).



Decomposability: Breaking Problems Apart

DECOMPOSABLE

Symbolic Integration

$$\int x^2 + 3x + \sin^2 x \cdot \cos^2 x \, dx$$

Break into THREE independent sub-problems:

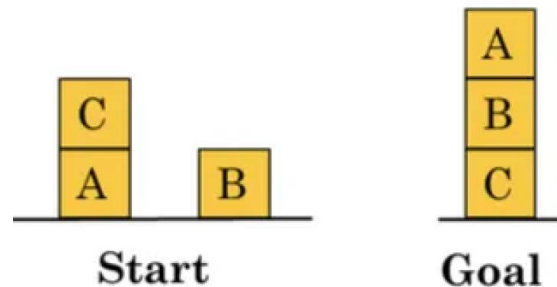
- $\int x^2 \, dx$ (polynomial rule)
- $\int 3x \, dx$ (constant $\times$ polynomial)
- $\int \sin^2 x \cdot \cos^2 x \, dx$ (trigonometric identity)

Solve each separately $\rightarrow$ combine results!

NON-DECOMPOSABLE

Blocks World Problem

Move blocks from Start $\rightarrow$ Goal configuration



Steps are INTERDEPENDENT — must move A before C can sit on top of B. Cannot decompose!



Solution Step Recoverability

A critical design decision: can wrong steps be undone, or must every decision be final?

Problems fall under three classes:

IGNORABLE

Example: Theorem Proving

A wrong step can be completely ignored. The proof still proceeds without any penalty. Control: simple, no backtracking needed.

RECOVERABLE

Example: 8-Puzzle / Sliding Tiles

A wrong move can be undone by backtracking — at the cost of extra steps. Control: allows backtracking in the search tree.

IRRECOVERABLE

Example: Chess / Card Games

A wrong move cannot be ignored or undone. Must maximize utility of the current state. Control: costly deliberate search before each decision.



Predictability & Solution Type

PREDICTABILITY: Is the problem universe deterministic or uncertain?

Certain Outcome

8-Puzzle, Water Jug — planning generates a guaranteed solution sequence. Planning avoids wasted steps.

Uncertain Outcome

Playing Cards — only probable solutions; search tree grows exponentially at each uncertainty point.

SOLUTION TYPE: Any acceptable path vs. the best possible path?

Any-Path Problem

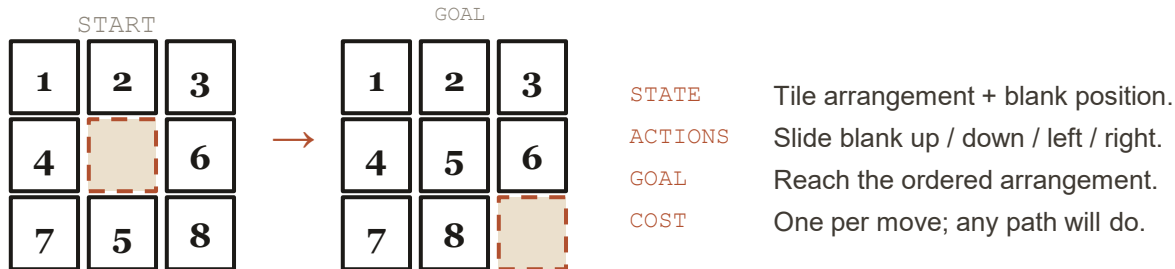
Water Jug, 8-Puzzle — any valid solution suffices. Heuristics guide to first acceptable solution quickly.

Best-Path Problem

Travelling Salesman — must find the SHORTEST route. Requires exhaustive or near-exhaustive search. Computationally harder!

1. The 8-Puzzle

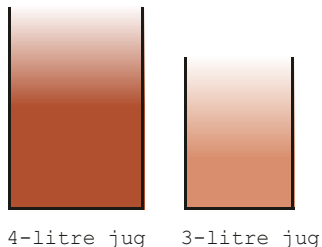
EXAMPLES



Scale — the 3×3 puzzle has $9!/2 = 181,440$ reachable states; the 15-puzzle has over 10^{13} . This explosion is exactly why heuristics (e.g. misplaced tiles, Manhattan distance) are needed.

2. The Water Jug Problem

Given a 4-litre and a 3-litre jug and a tap, measure out exactly **2 litres**. The *solution is the path* — the sequence of operations — not merely the final state.



Fill a jug

Empty a jug

Pour A → B

Pour B → A

A state is a pair (x, y) = litres in each jug. Goal: any state with $x = 2$.

One solution — $(0,0) \rightarrow \text{fill 4} \rightarrow (4,0) \rightarrow \text{pour into 3} \rightarrow (1,3) \rightarrow \text{empty 3} \rightarrow (1,0) \rightarrow \text{pour} \rightarrow (0,1) \rightarrow \text{fill 4} \rightarrow (4,1) \rightarrow \text{pour into 3} \rightarrow (2,3)$. Goal reached.

Two Families of Search

UNINFORMED · BLIND

No domain knowledge

Explores using only the problem definition — cannot tell a promising state from a poor one.

BFS

DFS

UCS

DLS

IDS

INFORMED · HEURISTIC

Guided by an estimate

Uses a heuristic $h(n)$ — estimated distance to the goal — to expand the most promising states first.

Greedy best-first

A* search

The trade — uninformed search is general but blind and often slow; informed search is faster but needs a domain-specific heuristic. A good heuristic is the difference between intractable and instant.

SEARCH STRATEGIES

Five Uninformed Strategies

They differ only in *the order* in which they expand nodes.

Breadth-First (BFS)

Expand shallowest node first — level by level.

Depth-First (DFS)

Plunge down one branch before backtracking.

Uniform-Cost (UCS)

Expand the lowest path-cost node first.

Depth-Limited (DLS)

DFS with a maximum depth cap.

Iterative Deepening (IDS)

Repeated DLS with a growing limit — DFS memory, BFS completeness.

System Characteristics

Properties that describe an AI system's design and behavior



Performance Measure

Criteria that define success. What does it mean for the agent to succeed? E.g., maximize score, minimize cost, complete a task.



Environment

The external world the agent perceives and acts upon. Can be fully/partially observable, static/dynamic, discrete/continuous.



Actuators

The mechanisms through which the agent acts on the environment. E.g., motors, display outputs, network requests.



Sensors

The mechanisms through which the agent perceives the environment. E.g., cameras, microphones, APIs.

PEAS = Performance • Environment • Actuators • Sensors

SECTION 04

Intelligent Agents

What agents are, how they work, and how they are classified

WHAT IS AN AGENT?

01

SENSE

Perceives its environment
via sensors

02

PERCEIVE

Processes and interprets
incoming inputs

03

REASON

Evaluates and decides on a
response

04

ACT

Executes actions via
actuators

Example: Thermostat

Senses temperature → Compares to threshold → Decides to heat/cool → Turns heater on/off.

Note: Meets agent criteria but is NOT intelligent — it uses fixed if-else rules with no learning.



What is an Intelligent Agent?

An intelligent agent is an autonomous entity that perceives its environment via sensors and acts via actuators to achieve goals

Formal Definition: Agent: Percepts* → Actions

Percepts = complete history of all the agent has ever perceived Actions = the chosen action at this moment in time*

Example: Self-Driving Car Agent

SENSORS

Cameras · Radar · GPS · LiDAR

ACTUATORS

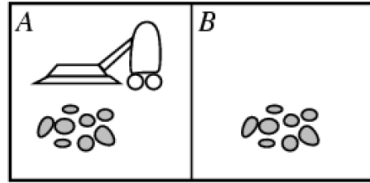
Steering Wheel · Brakes · Throttle

GOAL

Navigate safely from A → B obeying all traffic laws

"The functional view provides a powerful, abstract way to model any AI system, regardless of internal complexity."

Vacuum-cleaner world



Percepts: location and contents, e.g., [A,Dirty]

Actions: *Left*, *Right*, *Suck*, *NoOp*

A vacuum-cleaner agent

Percept sequence	Action
[A,Clean]	<i>Right</i>
[A,Dirty]	<i>Suck</i>
[B,Clean]	<i>Left</i>
[B,Dirty]	<i>Suck</i>
[A,Clean], [A,Clean]	<i>Right</i>
[A,Clean], [A,Dirty]	<i>Suck</i>
...	...

What is the **right** function?

Can it be implemented in a small agent program?

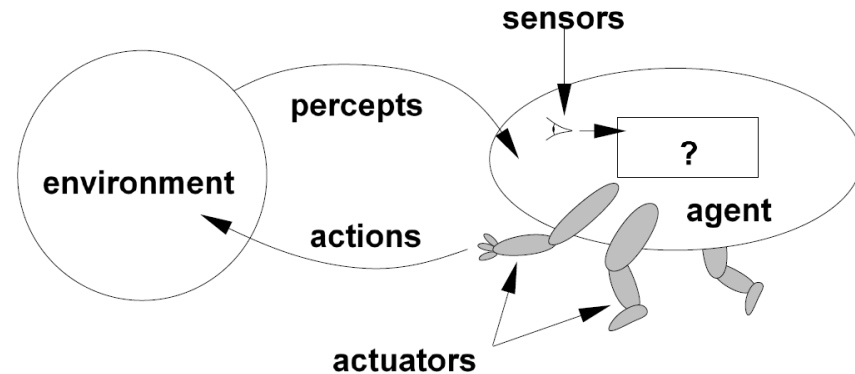
Agents include humans, robots, softbots, thermostats, etc.

The **agent function** maps from percept histories to actions:

$$f: P^* \rightarrow A$$

The **agent program** runs on the physical **architecture** to produce f .

Agents and environments





The Agent-Environment Interaction Model

A continuous, unbroken loop forming the basis of all autonomous AI systems:



PERCEIVE

Sensors gather raw data from the environment



THINK

Internal architecture processes data and selects action



ACT

Actuators execute the chosen action on the environment

The Percept-Think-Act Loop

This cycle runs continuously from activation to shutdown — from a chatbot responding to a user message, to a factory robot picking up components. The model's elegance is its universality.



Properties of Rational Agents

A rational agent does whatever action maximizes its performance measure, given its percept sequence and internal knowledge.

Rationality $\neq$ Omniscience — the agent makes the **BEST** possible decision with **AVAILABLE** information



Performance Measure

The criterion that defines success. For a vacuum robot: clean floor per hour. For chess: winning.



Percept Sequence

Complete history of everything perceived up to now — the agent's experiential memory.



Prior Knowledge

Built-in model of how the world works — e.g., braking makes a car slow down.



Available Actions

Agent cannot perform actions beyond its capabilities, no matter how rational it is.



Agent Types

1

Simple Reflex

Current percept only

Example: Thermostat

Limitation: Fails in partial observability

2

Model-Based Reflex

Internal world model

Example: Roomba with map

Limitation: No explicit goal

3

Goal-Based

Model + explicit goal

Example: Google Maps navigation

Limitation: No preference between goal paths

4

Utility-Based

Goal + utility function

Example: Stock trading bot

Limitation: Needs a utility measure defined

5

Learning Agent

Adapts from experience

Example: Netflix recommender

Limitation: Requires sufficient training data



Types of Agents: 1. Simple Reflex Agents



Simple Reflex Agent

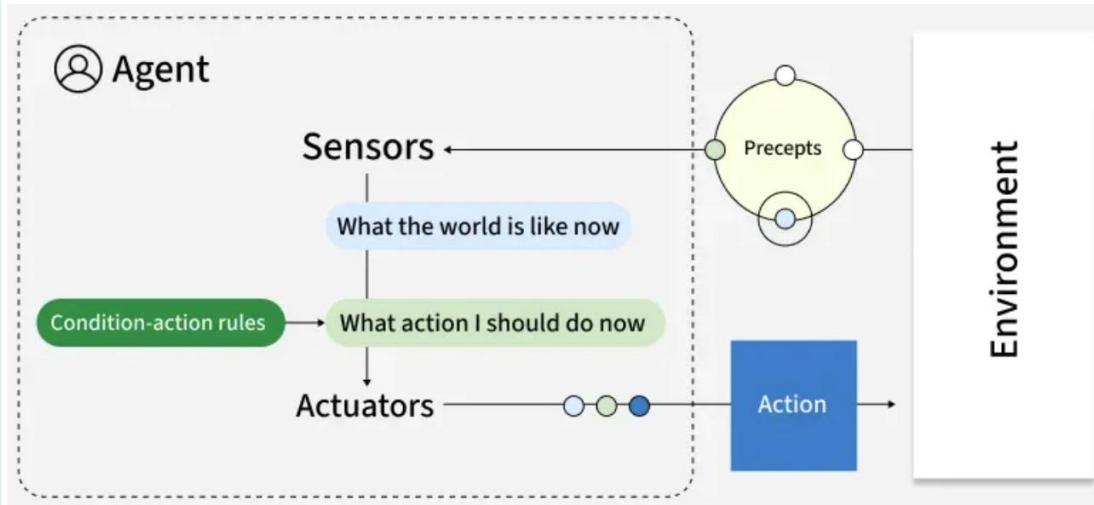
Acts on CURRENT percept only — no memory of past

Condition-Action Rule:
IF [condition] THEN [action]

Example: Thermostat

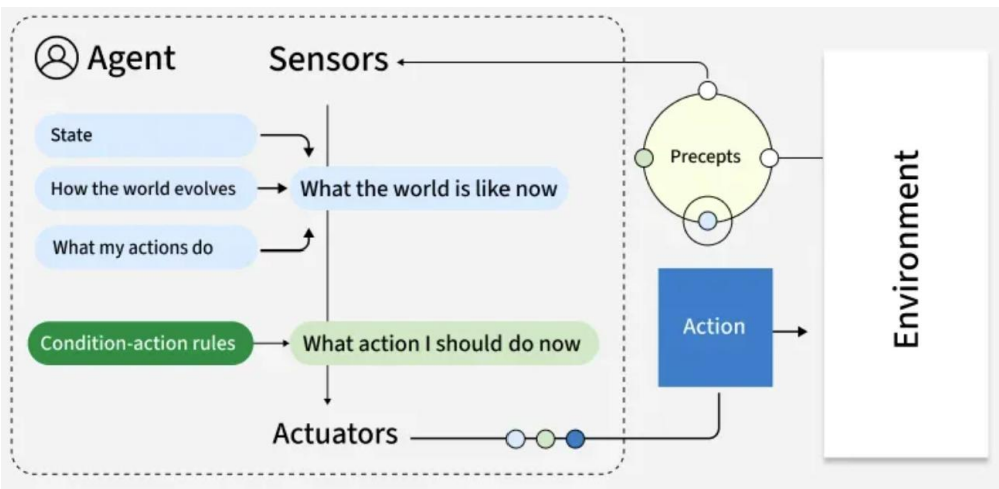
- Perceives: current temperature
- If temp < set point → turn on heater
- No memory of how long it's been cold
- No planning — pure reaction

Simple reflex agents act only on the current perception of the environment using predefined condition–action rules. They do not rely on past experiences or predict future outcomes and respond directly using simple “if–then” logic.





2. Model-Based Reflex Agents



- Model-based reflex agents maintain an internal model of the environment to handle situations where full information is not directly available.
- This helps them make better decisions by considering changes in the environment and the impact of their actions.



Model-Based Reflex Agent

Maintains internal model of world — handles partial observability

Tracks what it cannot currently see.

Example: Roomba Vacuum Robot

- Builds an internal MAP of rooms
- Knows which rooms are already cleaned
- Does not revisit clean areas
- Acts more efficiently — avoids redundant work



3. Goal-Based Agents



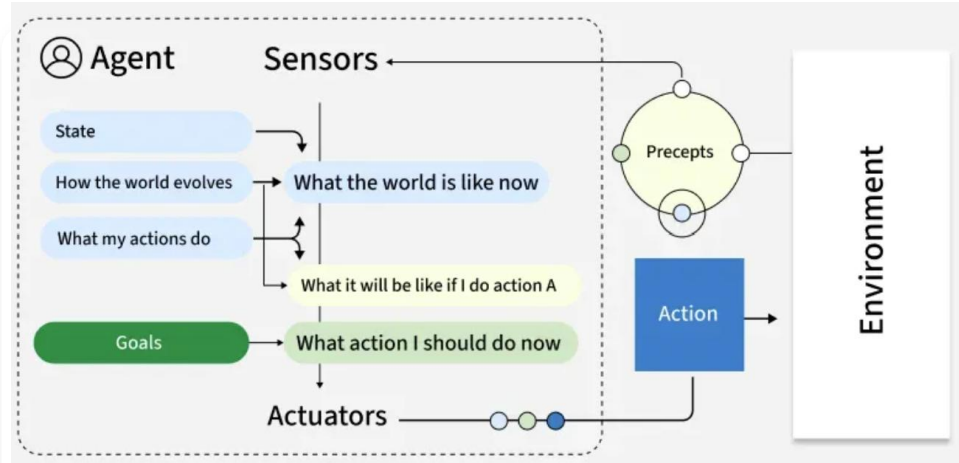
Goal-Based Agent

Internal model + explicit goal → considers future consequences

Key question: "Will this action help me reach my goal?"

Example: Google Maps Navigation

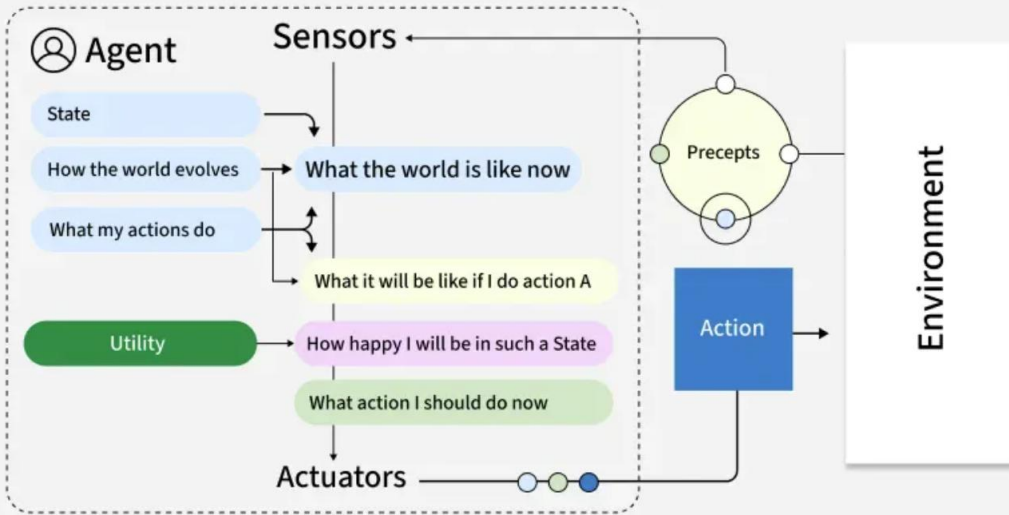
- Perceives: current location + destination
- Plans: sequence of turns to reach goal
- Considers: road options, turns, distances
- Selects best path to achieve the objective



- Goal-based agents choose their actions by focusing on a specific objective and evaluating how different choices can help achieve it.
- Instead of reacting only to the current situation, they plan ahead and consider possible future outcomes.



4. Utility-Based Agents



- Utility-based agents go beyond simply achieving goals by evaluating how beneficial each action is using a utility function, which measures the overall “value” or satisfaction of an outcome.
- This helps them choose the best option when dealing with trade-offs or uncertainty.



Utility-Based Agent

Multiple goal paths — chooses action that maximizes utility

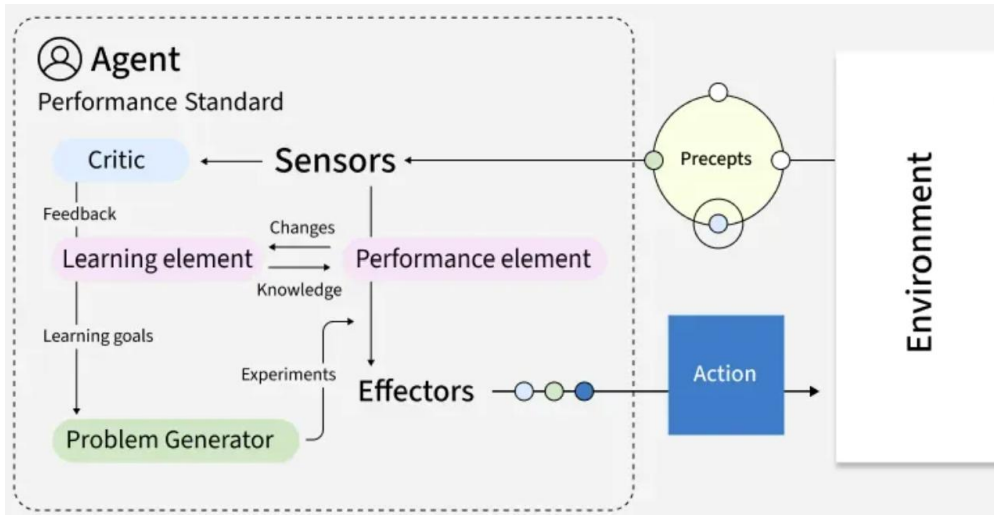
Utility = measure of happiness / performance / profit

Example: Stock Trading Bot

- Goal: make profit
- But: not any profitable trade — the ONE expected to yield the HIGHEST profit
- Balances potential gain vs. risk of loss
- Makes decisions that maximize expected utility



5. Learning Agents



Learning agents improve their behavior over time by using feedback from past actions.

They continuously refine their internal models to make better decisions in future situations.



Learning Agents

- Adapt their behavior based on experience and feedback
- Improve decision-making by updating internal models
- Combine action execution with a learning mechanism
- Create new knowledge instead of relying only on fixed rules



The PEAS Framework for Specifying Agents

PEAS = Performance Measure | Environment | Actuators | Sensors

Agent	P — Performance	E — Environment	A — Actuators	S — Sensors
Medical Diagnosis System	Patient health outcomes; correct diagnoses; minimized side effects	Patient data, symptom reports, test results, medical records	Treatment recommendations, test requests, referrals	Symptom inputs, lab results, patient history database
Self-Driving Taxi	Safe arrival, legal compliance, passenger comfort, trip time	Roads, traffic, pedestrians, weather, GPS maps, other vehicles	Steering, brakes, throttle, indicators, horn	Cameras, lidar, radar, GPS, speedometer, ultrasonic sensors
Interactive English Tutor	Student test scores, learning speed, engagement retention	Student answers, keyboard inputs, school curriculum	Display exercises, give feedback, suggest resources	Student responses, typed text, time-on-task metrics

AI Agent

An **AI agent** is the *practical implementation term*, especially in modern AI systems.

It usually implies:

- Uses AI techniques (LLMs, machine learning, planning)
- Can **reason, plan, use tools, and sometimes act autonomously**
- Often part of systems like:
 - LLM-based agents (ChatGPT-style tool-using agents)
 - Autonomous workflow agents (coding agents, web agents)

Examples:

- A ChatGPT-based agent that can browse, write code, and execute tasks
- AutoGPT-style systems
- Customer support bots with tool access

Key difference

Aspect	Intelligent Agent	AI Agent
Term origin	Classical AI theory	Modern applied AI
Scope	Broad, formal definition	Practical implementations
Must use ML/LLMs?	No	Often yes
Focus	"What an agent is"	"What an AI system does"

MODERN LLM-BASED IMPLEMENTATIONS

Anthropic · Google · LangChain

WORKFLOWS

(Anthropic definition)

Prompt Chaining — Sequential task decomposition

Routing — Classifies & directs to relevant tools

Parallelization — Concurrent subtask execution

Orchestrator-Workers — Central LLM delegates subtasks

Evaluator-Optimizer — Iterative produce & evaluate loop

AGENTS

(Anthropic definition)

LLMs manage their own internal processes and tool usage independently.

Recommended for open-ended tasks where steps can't be predicted or hardcoded.

Google's Agent Components

1. Orchestrator
2. Model (LLM)
3. Tools

⚠ Most current implementations are closer to workflows or reflex agents — not truly intelligent agents.

SECTION 05

Environments & Agent Structures

Classifying environments and designing agent architectures



Environment Properties: Six Key Dimensions

The nature of the environment profoundly shapes which agent architecture is appropriate:

Fully vs Partially Observable

Full: agent sees complete state (Chess board)

Partial: limited view (fog, blind spots)

Deterministic vs Stochastic

Deterministic: same action → same outcome (puzzle)

Stochastic: random element (stock market, dice)

Episodic vs Sequential

Episodic: each episode independent

Sequential: current decision affects future (robot path planning)

Static vs Dynamic

Static: world unchanged while agent thinks (crossword)

Dynamic: changes while deliberating (soccer robot)

Discrete vs Continuous

Discrete: finite states and actions (tic-tac-toe)

Continuous: infinite smooth states (drone flight)

Single-Agent vs Multi-Agent

Single: one intelligent agent (crossword solver)

Multi-agent: multiple agents interact (chess, traffic)



Observable vs Partially Observable Environments



Fully Observable



Agent's sensors access the COMPLETE state of the environment at each moment

Example: Chess Game

- Entire board is always visible
- Both players see all pieces
- No hidden information
- Agent can plan with complete information

Design advantage: simpler agent — no need to model unobserved state



Partially Observable

Agent only sees LIMITED information — must infer or model the rest

Example: Self-Driving Car

- Cannot see behind buildings
- Cannot see what's in other cars' blind spots
- Sensor noise and occlusion

Design requirement: must maintain internal model (belief state) to estimate unseen world state



Deterministic vs Stochastic Environments



Deterministic

Next state completely determined by current state + action

Example: Sliding Tile Puzzle

- Moving tile left always gives same result
- No randomness involved
- Agent can plan entire solution in advance
- No need for probabilistic reasoning

Design: simpler planning algorithms suffice



Stochastic

Same action in same state can lead to different outcomes

Example: Stock Market

- Same trading action today → different profit tomorrow
- Unpredictable market forces
- Requires probabilistic models
- Must plan for multiple contingencies

Design: needs probability distributions, not deterministic paths



Static/Dynamic & Discrete/Continuous Environments

Static vs Dynamic

Static

World unchanged while agent deliberates
Example: Crossword puzzle — clues don't change

Dynamic

World changes while agent thinks
Example: Soccer robot — ball moves, opponents react!

Discrete vs Continuous

Discrete

Finite, countable states and actions
Example: Tic-tac-toe — limited board configurations
Agent can enumerate all possibilities

Continuous

Infinite, smoothly varying states and actions
Example: Drone navigating the sky — unlimited positions and movements. Requires calculus-based methods.



Case Study: AI in Autonomous Vehicles

An autonomous vehicle is not one agent — it is a SYSTEM of specialized agents:

Environment: Dynamic · Partially Observable · Stochastic · Continuous · Multi-Agent

Perception Agent

Learning-Based

Processes camera & LiDAR data to detect lane lines, traffic signs, pedestrians, and other vehicles.

Decision Agent

Utility-Based

Makes real-time decisions: accelerate, brake, or change lanes? Balances speed, safety, and comfort utility.

Planning Agent

Goal-Based

Calculates optimal route to destination on a longer time horizon — like an enhanced GPS with context.

Control Agent

Optimization

Precise motion control — translates decisions into exact steering angles, brake pressures, and throttle positions.

A self-driving vehicle beautifully illustrates how AI is a hybrid discipline: search + reasoning + probability + learning + optimization working in concert.

SECTION 06

Ethics in AI

Navigating responsibility in the age of intelligent machines



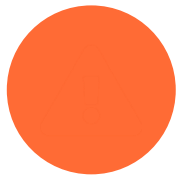
Why AI Ethics Matters

"Understanding and mitigating risks is as crucial as understanding the technology itself." — Akintayo, 2025



Scale of Impact

AI systems make millions of decisions per day affecting healthcare, finance, criminal justice, and hiring — errors are amplified at scale.



Opacity

Many deep learning models are 'black boxes' — even their creators cannot fully explain why specific decisions were made.



Power Asymmetry

Those who build and deploy AI have enormous power over those affected by it — creating new ethical responsibilities.



Data Privacy & Surveillance

AI systems are trained on — and operate using — vast amounts of personal data:

1

Personal Data at Scale

Social media, browsing history, location data, medical records, financial transactions — all used to train AI systems, often without explicit awareness.

2

Facial Recognition Risks

Mass deployment of facial recognition in public spaces enables unprecedented surveillance. Misidentification rates are significantly higher for darker-skinned individuals (MIT Media Lab, 2018).

3

Data Consent

Users often unknowingly grant broad data rights via terms-of-service agreements. The 'cost' of free services is personal data — a trade many don't fully understand.

4

Cross-Context Profiling

Data from one context (health app) may be combined with another (social media) to build profiles users never consented to — violating contextual integrity.

Regulatory Response: GDPR (EU), CCPA (California), PIPL (China) — frameworks emerging globally to protect personal data



AI Bias, Fairness & Accountability

Bias in → Bias out: AI trained on historical data inherits and amplifies human prejudices

Historical Bias

Hiring AI trained on male-dominated tech industry resumes down-ranks female applicants — Amazon's scrapped AI recruiter (2018).

Measurement Bias

Facial recognition misidentifies Black and Asian faces at much higher rates than white faces — systemic measurement inequity.

Representation Bias

Healthcare AI trained predominantly on data from one demographic group performs worse for underrepresented patients.

Confirmation Bias

Recommendation algorithms create echo chambers — reinforcing existing beliefs rather than exposing users to diverse perspectives.

Accountability: When AI makes a consequential error — who is responsible? Developer · Deployer · User · Regulator?



Principles for Ethical AI Development

Transparency

AI decisions should be explainable — users deserve to understand why a system made a specific choice affecting them.

Fairness

Systems must treat all groups equitably; disparate impact across demographic groups should be detected and corrected.

Privacy

Collect only necessary data; obtain meaningful consent; protect personal information throughout the AI lifecycle.

Accountability

Clear lines of responsibility for AI outcomes — developers, deployers, and users all have obligations.

Safety & Robustness

Systems must behave reliably in edge cases and resist adversarial attacks, with human oversight maintained.

Human Autonomy

AI should augment human capability, not replace human judgment in high-stakes decisions affecting people's lives.